



Chemical & Water Desalination Engineering Program

Study Plan Booklet

Academic Year 2023-2024

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Chemical & Water Desalination Engineering Program
Academic Year 2023-2024

Semester-by-semester Study Plan
(131 Total Credit Hours - 52 Total Courses)

University Requirement	(24 Credit Hours)
College Requirement	(26 Credit Hours)
Program Requirement	(81 Credit Hours)

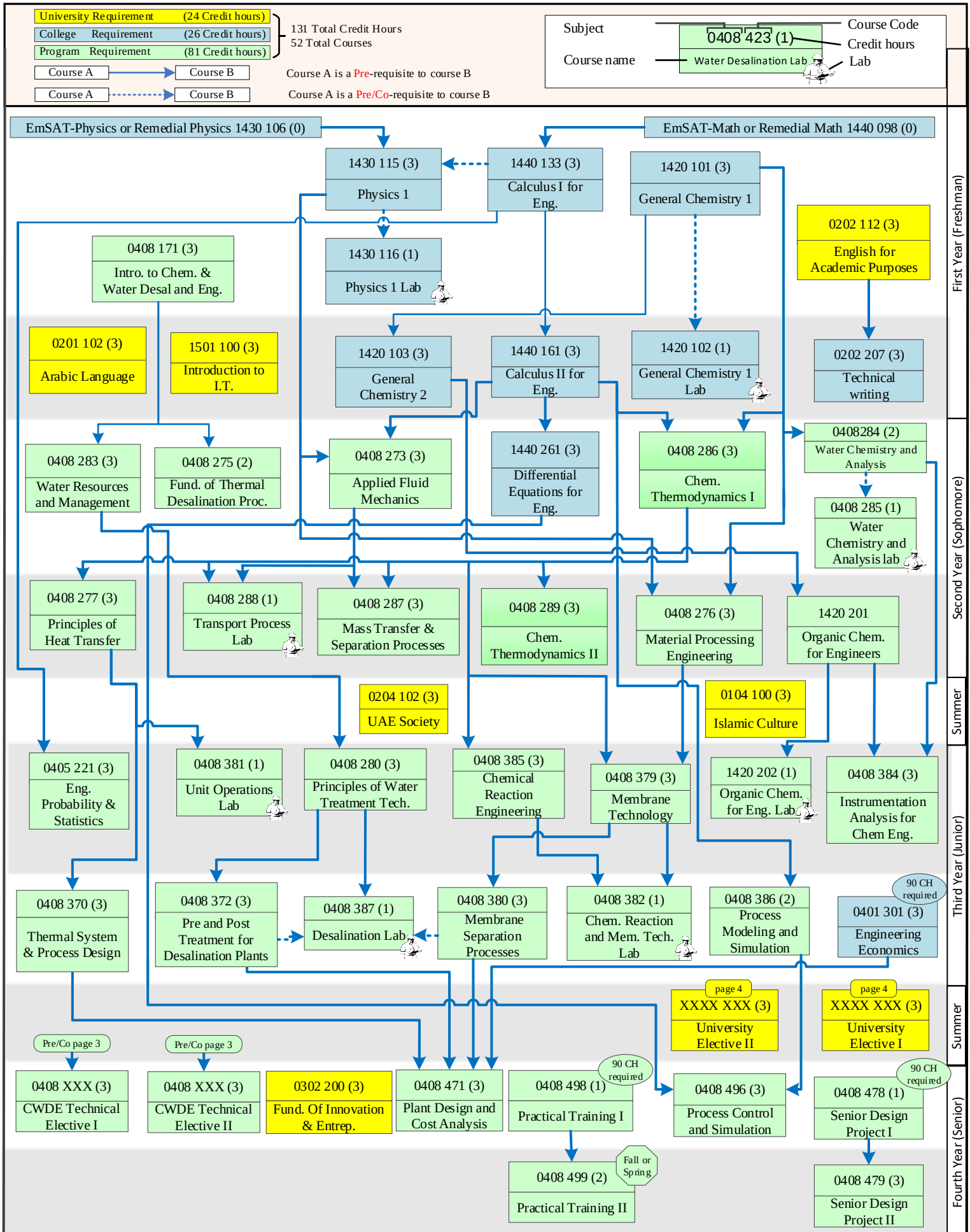
Course Number Breakdown				
Mass Transfer	College Code	Department Code	Level	Course Code
0408 278	04	08	2	78

First Year (Freshman Year)				
First Semester			Second Semester	
0202 112	English for Academic Purposes	3	0202 102	Arabic Language 3
1430 115	Physics I	3	1501 100	Introduction to I.T. 3
1430 116	Physics I Lab	1	1440 161	Calculus II for Engineering 3
1420 101	General Chemistry (1)	3	0202 207	Technical Writing 3
1440 133	Calculus I for Engineering	3	1420 103	General Chemistry 2 3
0408 xxx	Intro to Chem and Water Desal Eng	3	1420 102	General Chemistry 1 Lab 1
	Total Credit Hours	16		Total Credit Hours 16
Second Year (Sophomore Year)				
Third Semester			Fourth Semester	
0408 283	Water Resources and Management	2	0408 277	Principles of Heat Transfer 3
0408 284	Water Chemistry and Analysis	2	0408 287	Mass Transfer & Sep. Proc. 3
0408 285	Water Chemistry Analy. Lab	1	1420 201	Organic Chem. For Engineers 3
0408 286	Chemical Thermodynamics 1	3	0408 289	Chemical Thermodynamics 2 3
0408 273	Applied Fluid Mechanics	3	0408 288	Transport Processes Lab 1
0408 275	Fund. Of Therm Desal. Proc.	2	0408 276	Material Process Engineering 3
1440 261	Diff. Equations for Eng	3		
	Total Credit Hours	16		Total Credit Hours 16
Summer Semester				
0104 100	Islamic Culture	3		
0204 102	UAE Society	3		
Third Year (Junior Year)				
Fifth Semester			Sixth Semester	
0408 384	Instru. Analy for Chem. Engr.	3	0401 301	Engineering Economics 3
0408 280	Princ. Of Water Treat. Tech	3	0408 386	Process Modelling & Simul 2
0408 379	Membrane Technology	3	0408 380	Membrane Separation Proc. 3
0405 221	Eng. Probability & Statistics	3	0408 382	Chem React & Mem Tech Lab 1
0408 385	Chemical Reaction Eng.	3	0408 370	Thermal Sys & Process Design 3
0408 381	Unit Operations Lab	1	0408 372	Pre-Post Treat, for Desal. Plant 3
1420 202	Organic Chem. For Eng. Lab	1	0408 387	Desalination Laboratory 1
	Total Credit Hours	17		Total Credit Hours 16
Summer Semester				
xxxx xxx	University Elective I	3		
xxxx xxx	University Elective II	3		
Fourth Year (Senior Year)				
Seventh Semester			Eighth Semester	
0408 496	Process Control and Simulation	3	0408 479	Senior Design Project II 3
0408 471	Plant Design and Cost Analysis	3	0408 499	Practical Training 2 2
0480 478	Senior Design Project I	1		
0408 498	Practical Training I	1		
0302 200	Fund. of Innovation & Entrepreneur	3		
0408 xxx	CWDE Technical Elective II	3		
0408 xxx	CWDE Technical Elective II	3		
	Total Credit Hours	17		Total Credit Hours 5
TOTAL CREDIT HOURS = 131				

Chemical & Water Desalination Engineering Program

Academic Year 2023-2024

Study Plan Flowchart



Chemical & Water Desalination Engineering Program
Academic Year 2023-2024
List of Technical Electives

#	Course Code	Course Name	Pre-requisite	Credit Hours
1	0408 480	Renewable Energy Desalination	None	3
2	0408 481	Hybrid Desalination Processes	None	
3	0408 482	Modeling & Simulation of Multi-Stage Flash Desalination	None	
4	0408 483	Membrane Distillation	None	
5	0408 484	Pressure Retarded Osmoses (PRO) Membranes	None	
6	0408 485	Nanotechnology for Water Treatments and Desalination	None	
7	0408 486	Nuclear Desalination	None	
8	0408 487	Special Topic	None	
9	0408 476	Water and Energy Nexus	None	
10	0408 489	Emerging Membrane Processes	None	
11	0408 493	Introduction to Biochemical Engineering	None	
12	0408 495	Process Safety	None	
13	0408 494	Oil and Gas Technology	None	

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Academic Year 2023-2024
List of University Electives

Important Notes:

You must select one course only from each of the below 2 domains of courses. If you select two courses from the same group, only one course will be considered for you.

Domain 1 (Humanities Social and Arts)			
#	Course Code	Course Name	Credit Hours
1	0103 103	Islamic System	3
2	0103 104	Prof. Ethics in Islamic Sharia	
3	0104 130	Analytical Biog of the Prophet	
4	0201 140	Intro. to Arabic Literature	
5	0202 130	French Language	
6	0203 100	Islamic Civilization	
7	0203 102	History of the Arabian Gulf	
8	0203 200	Hist of Sciences among Muslims	
9	0204 103	Principles of Sign Language	
10	0206 102	Fundamentals/Islamic Education	
11	0206 103	Introduction to Psychology	
12	0302 150	Intro.to Bus for Non-Bus	
13	0308 131	Personal Finance	
14	0308 150	Intro to Economics (Non-B)	
15	0602 246	Human Rights in Islam	
16	0700 310	Intro to Islamic Art & Design	
17	0710 109	Arts and Medicine	
18	0800 107	Media in Modern Societies	
19	0900 107	History of Medical and H.Sc.	

Domain 2 (Natural & Health Sciences)			
#	Course Code	Course Name	Credit Hours
1	0401 142	Man and The Environment	3
2	0406 102	Introduction to Sustainability	
3	0503 101	Health and Safety	
4	0505 101	Fitness and Wellness	
5	0507 101	Health Awareness and Nutrition	
6	1430 101	Astro & Space Sciences	
7	1450 100	Biology and Society	
8	1502 233	Introduction to AI	